SOLSTICE OPAL HOTEL

ANCILLARY REVENUE

OPPORTUNITY REPORT

Who generates ancillary revenue, what the purchasing patterns reveal, and which opportunities the hotel should test next.

$60,620
Recorded ancillary revenue

349 / 400
Guests with recorded spend

87.25%
Recorded participation

Based on the DataCamp sample hotel datasets and the linked Python notebook. Figures are descriptive unless marked as illustrative scenarios.

DATA ANALYST CERTIFICATION • SAMPLE PRACTICAL EXAM

EXECUTIVE SUMMARY

The business story

A concise reading of the data: revenue is concentrated in large guest segments, while higher loyalty tiers show greater spend per purchasing guest.

Central finding. Leisure guests with no loyalty tier are the largest identifiable combined segment by total recorded ancillary revenue ($10,901). Platinum leisure guests have the highest average total spend per purchasing guest ($404).

WHO CONTRIBUTES

Leisure travellers represent 57.5% of revenue with known travel purpose, compared with 42.5% from business travellers.

WHERE TO TEST

51 profiled guests have no recorded ancillary purchases; 42 of them have marketing consent.

HOW TO ACT

Pilot consent-based ancillary offers; measure first purchases and incremental revenue before scaling.

Interpretation guardrail: The loyalty relationship is statistically significant but weak. It does not establish that joining a loyalty programme causes guests to spend more.

How the analysis was built

The three source systems have different grains. Guest-level aggregation is essential to avoid overstating revenue.

Guest profiles

400 guest profiles

guest_id • loyalty_tier • marketing_consent

Stay details

600 recorded stays

guest_id • stay_id • travel purpose • booking channel • party size

Ancillary spend

800 source rows

guest_id • category • amount_spent

VALIDATED ANALYSIS RULES

1. Ancillary transactions were summed at guest level before joining profile attributes. 2. Stay purpose was reduced to one value per guest after confirming no conflicting business/leisure classifications. 3. Spending was not attributed to individual stays or booking channels.

IMPORTANT CLEANING LIMITATIONS

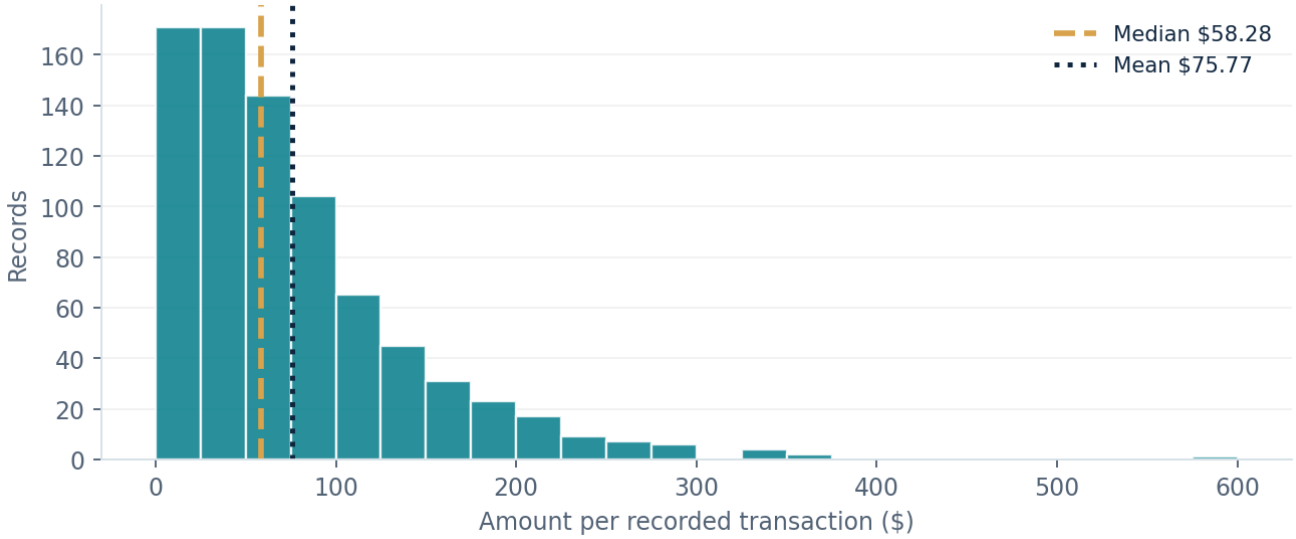
The original spending CSV contains 79 missing amount values. The notebook’s transaction distribution is consistent with these missing values being represented as zero; the BigQuery cleaning SQL is not included in the notebook, so the exact rule cannot be independently established. Treating a missing monetary value as zero is a substantive assumption.

Original profiles have 193 blank loyalty tiers, represented as “None” in this analysis. Guest IDs were standardised across systems. The 8.36% of revenue without a matched travel purpose remains unclassified rather than being assigned a category.

EXPLORATORY ANALYSIS

Transaction amounts are right-skewed

Distribution statistics are at transaction-record level, not total spend per guest.



$75.77
Mean transaction amount

$58.29
Median transaction amount

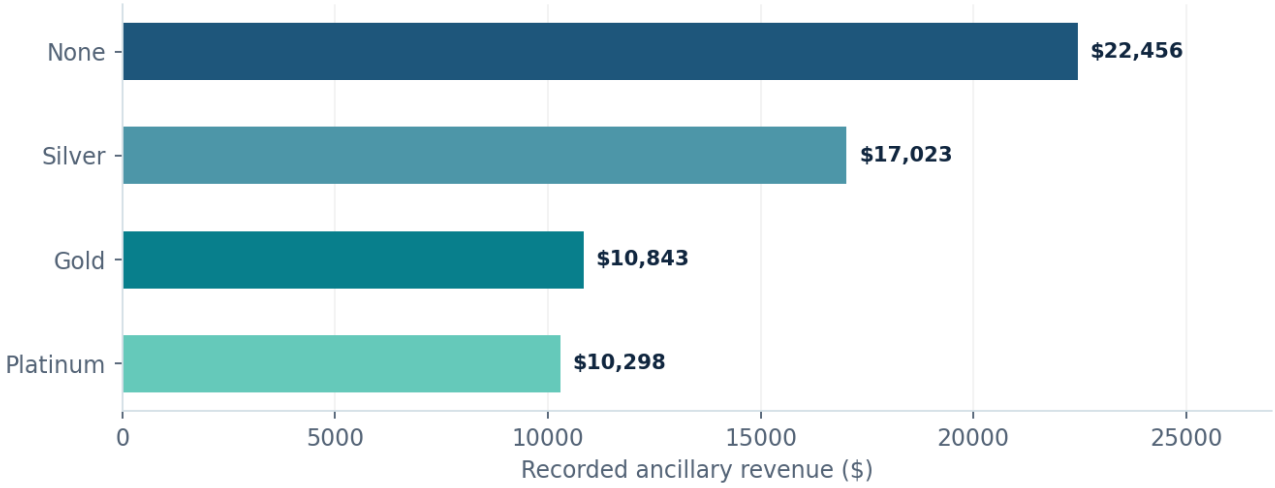
$104.83
75th percentile

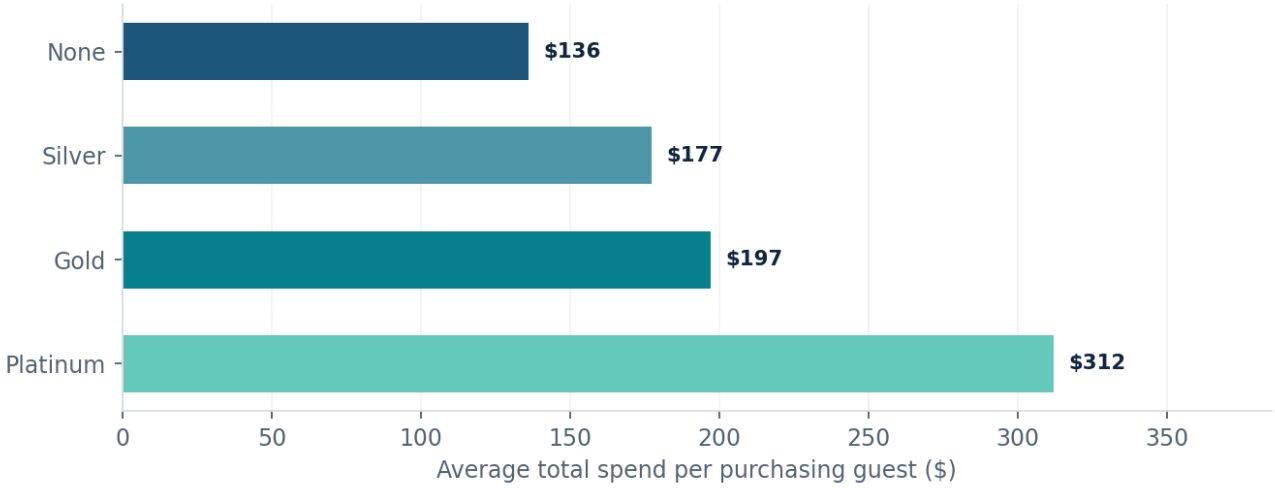
The mean exceeds the median and large purchases create a long right tail. The highest recorded amount is $578.90. Unusual high spending should not automatically be excluded: legitimate large purchases contribute to hotel revenue.

Data caveat: These statistics match the notebook's 800-record calculation that appears to include missing transaction amounts as zeros. For observed non-missing amounts alone, the median is $65.74 (721 records).

Loyalty: volume versus intensity

Guests without a recorded tier contribute the most total revenue; Platinum purchasers generate the highest average spend.





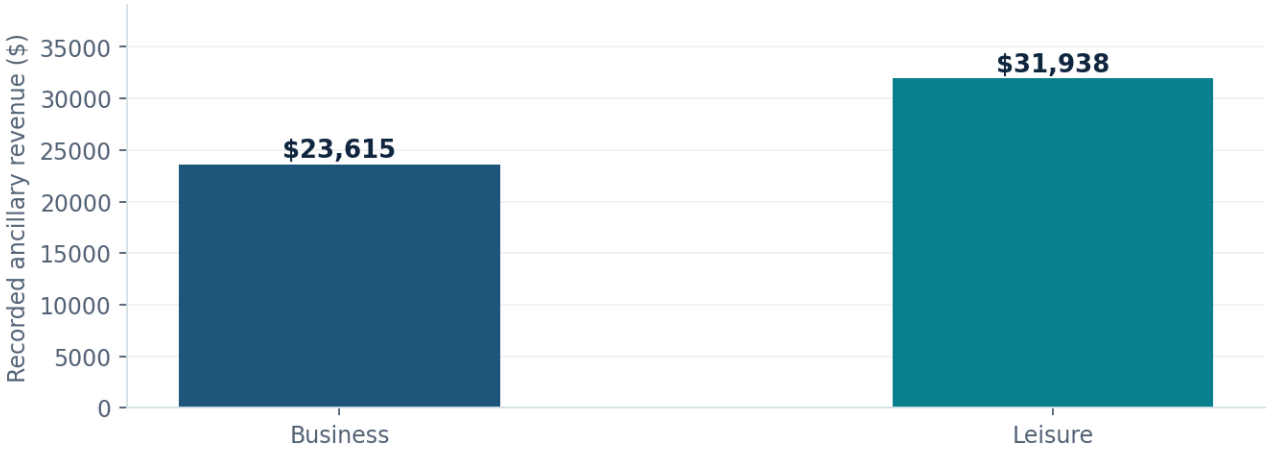
Volume effect: No-tier purchasers (165 guests) account for $22,456, or roughly 37.0% of total ancillary revenue. Intensity effect: Platinum purchasers (33 guests) average $312 in total recorded spend, versus $136 for no-tier purchasers.

A high segment total does not imply high spend per guest. The comparison is among purchasing guests and is not normalised by number of stays.

GUEST SEGMENTATION

Leisure leads on two measures

Travel-purpose analysis includes only guests with a verified business or leisure classification.



57.5%

Leisure share of classified revenue

$211.51

Average leisure purchaser spend

$191.99

Average business purchaser spend

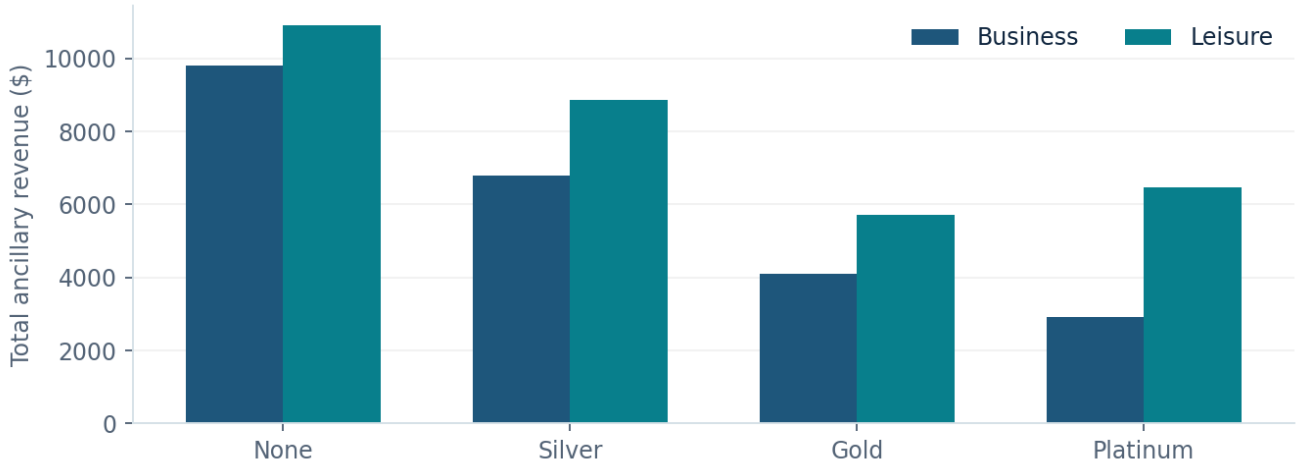
Leisure guests generate $31,938 from 151 purchasing guests; business travellers generate $23,615 from 123. Leisure has both a larger identifiable purchasing population and a moderately higher average spend per purchaser.

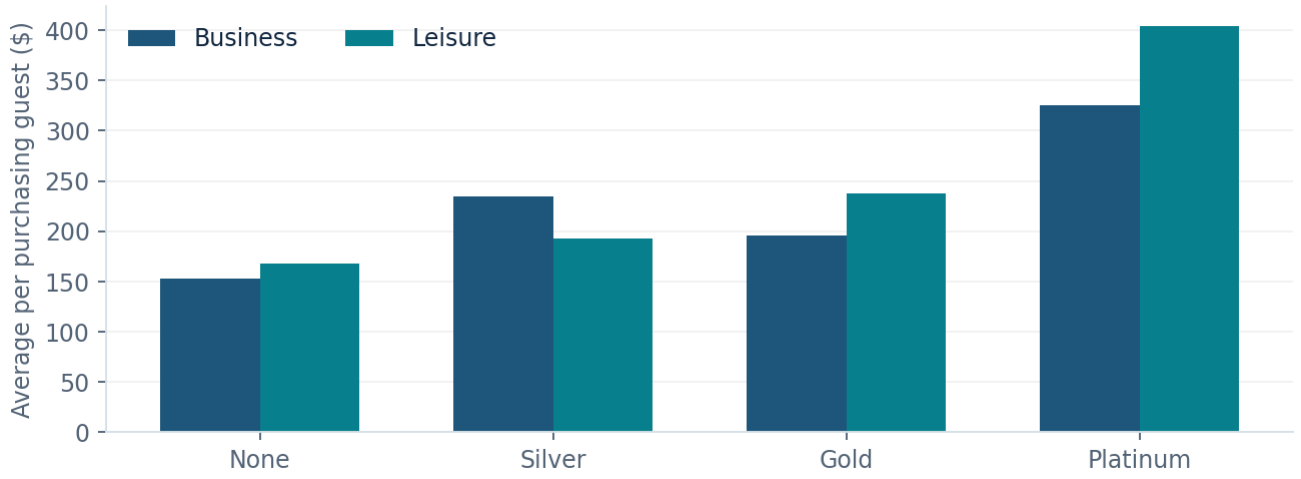
Revenue shares here use the $55,552 classified total. Another $5,067 (8.36% of hotel revenue) cannot be attributed to travel purpose.

COMBINED SEGMENTS

The leader depends on the metric

Crossing loyalty tier with travel purpose reveals distinct high-volume and high-spending groups.





Highest total: Leisure / None = $10,901 from 65 purchasing guests. Highest average: Leisure / Platinum = $403.79 per purchasing guest (16 guests). Business / Platinum follows at $325.07 (9 guests). Smaller segments warrant caution.

STATISTICAL EVIDENCE

A positive—but weak—loyalty association

The notebook applies a one-sided Spearman rank-correlation test to total guest-level spending among purchasers.

0.2265	9.65e-06	0.05
Spearman rho	One-sided p-value	Significance level

Test framing. Loyalty tiers were encoded None (0), Silver (1), Gold (2), and Platinum (3). The test evaluates whether higher ranked tiers tend to accompany higher recorded guest spending.

Conclusion. The p-value is below 0.05, indicating evidence against the null of no positive rank association. But rho = 0.2265 signals only a weak relationship. Neither the test nor the higher Platinum average establishes that loyalty enrolment itself increases spending.

WHAT MANAGEMENT CAN CLAIM

“Higher loyalty tiers tend to be associated with greater total recorded ancillary spending among purchasing guests.”

WHAT MANAGEMENT CANNOT CLAIM

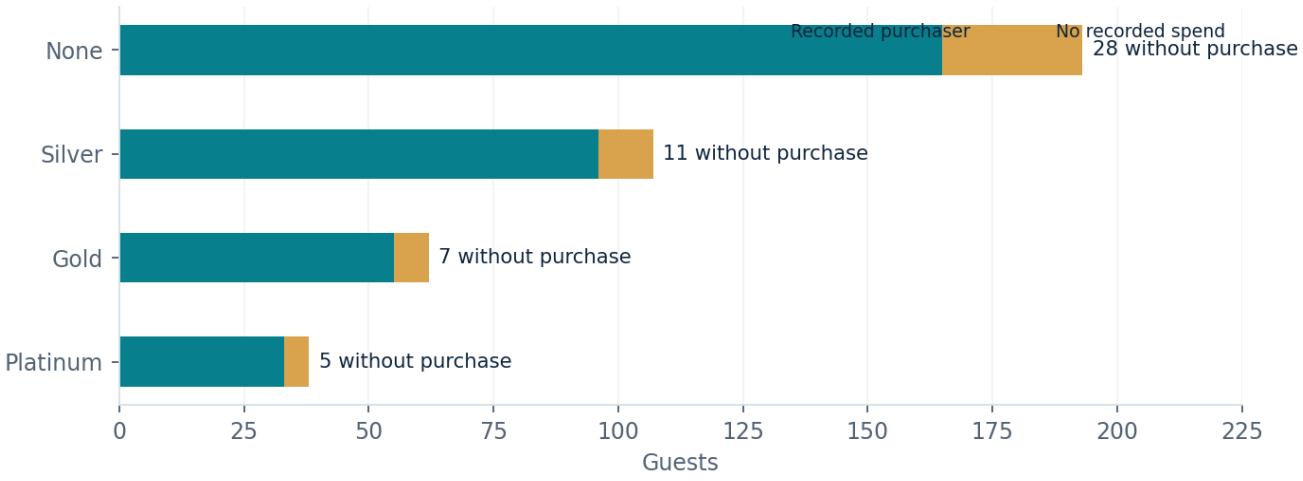
“Moving a guest into a higher tier will automatically make them spend more.”

The analysis is observational; possible differences in repeat visits or underlying guest preferences cannot be isolated because purchases do not map reliably to specific stays.

REVENUE OPPORTUNITY

51 guests have no recorded spend

Participation is high across the profiled guest base, but eligible non-purchasers provide a focused pilot audience.



400

Guest profiles

349

With recorded spending

87.25%

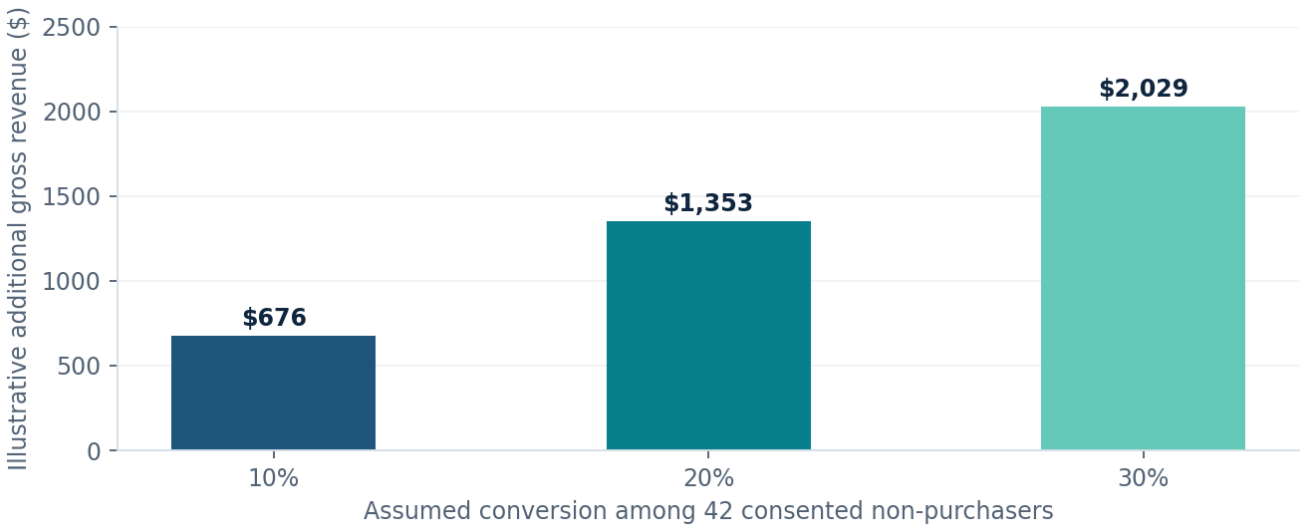
Participation baseline

Of the 51 profiles without recorded purchases, 28 are no-tier, 11 Silver, 7 Gold, and 5 Platinum. Forty-two of those 51 have marketing consent. Their absence from transaction records is treated as no observed purchase, not proof of zero lifetime spending.

Participation denominators must be explicit. The 87.25% figure is hotel-wide across 400 profiles; the 42-person targeted campaign requires its own prospective conversion metric.

Test small, then scale evidence

The campaign scenarios illustrate possible gross revenue under explicit assumptions; no campaign results are observed in this dataset.



Scenario assumption: 10%, 20%, or 30% of the 42 consented non-purchasers start buying, with expected spending equal to their loyalty tier's current average among purchasers. These imply approximately +1.12%, +2.23%, and +3.35% relative to $60,619.89 recorded revenue.

RECOMMENDED NEXT MOVES

1 — Targeted offer pilot. Test a relevant dining, spa, or activity offer for consented guests with no recorded purchase. 2 — Loyalty hypothesis. Explore enrolment incentives, but do not assume enrolment causes spending uplift. 3 — Measure incrementality. Compare treatment and control groups where feasible; account for costs before expansion.

KPI FRAMEWORK

Primary: Recorded ancillary purchase participation = 349 ÷ 400 = 87.25% (hotel baseline). For a pilot, track the fraction of eligible targeted guests who purchase during a fixed period. Secondary: incremental ancillary revenue attributable to the intervention versus control—not merely total participant sales.

Sources & limitations. DataCamp Solstice Opal Hotel brief; provided guest profiles, stays, and spending CSVs; GitHub notebook Datacamp_exam.ipynb (MuhammadUmar654/Datacamp_exam_sample). Missing amount treatment and original BigQuery cleaning steps cannot be fully audited from the notebook. No observed campaign outcomes; all uplifts are hypothetical and exclude costs.